

Machine Learning Advances in Photonic Systems: A Review of Methods and Applications

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Abstract: This paper reviews some representative applications of machine learning (ML) in photonics: inverse design of nanostructures, regression modeling of waveguides, and compact power splitter design. For each, the key challenge, ML approach, and limitations are mentioned to inform future development.

1. Introduction

Machine learning (ML) offers efficient alternatives to classical photonic design methods, especially where inverse problems or expensive numerical simulations are involved. This review summarizes how deep learning has been applied to inverse design and waveguide property modeling.

2. Reviewed Contributions

2.1 Inverse Design in Photonics via Deep Learning [1, 2]

Photonic design involves two fundamental challenges: the forward problem of calculating the optical response for a given structure and the more intricate inverse problem of designing a structure to achieve a desired response. Discriminative deep learning models, such as Convolutional Neural Networks (CNNs) for image-based structures and Fully Connected Networks (FCNs) for parameterized designs, are effectively used as surrogate physical models to predict optical responses.

The inverse problem is significantly more complex due to the vast, non-convex design space, the prevalence of local optima, and the degeneracy problem, where multiple structures can produce the same optical response. Deep learning addresses the inverse problem through various strategies. While methods using trained discriminative networks with iterative gradient-based optimization can find designs, they are primarily local optimization techniques susceptible to getting trapped in local minima. To better handle the one-to-many mapping and explore the solution space more effectively, deep generative models like Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) are employed. These models are adept at capturing the distribution of high-dimensional data and performing dimensionality reduction, encoding complex structures into a lower-dimensional latent space.

Beyond simply using generative models for sampling or dimensionality reduction, advanced hybrid approaches combine deep learning with traditional optimization for global optimization. A prominent example is the Global Topology Optimization Network (GLOnet). Unlike methods requiring a pre-compiled dataset of optimal designs, GLOnets are generative networks trained directly by incorporating feedback from physical simulations of the structures they generate. This physics-driven training allows the network to learn to generate globally optimal structures. Studies have shown that devices designed by GLOnets can achieve higher efficiencies than those found through solely local optimization methods, highlighting their potential to discover novel and high-performance photonic designs that are otherwise difficult to find. The diagram of the machine learning algorithm for photonic design is shown in Figure 1.

2.2 Regression Modeling of Nanophotonic Waveguides [3]

Designing waveguide metrics like effective index, power confinement, and coupling length is very important in photonics. These processes are done via some methods, such as FEM (Finite Element Method). But these methods are slow, time-consuming, and impractical for real-time optimization. In this paper, authors reduce time-consuming numerical simulations to design some waveguides, such as slot waveguides, strip waveguides, and directional couplers.

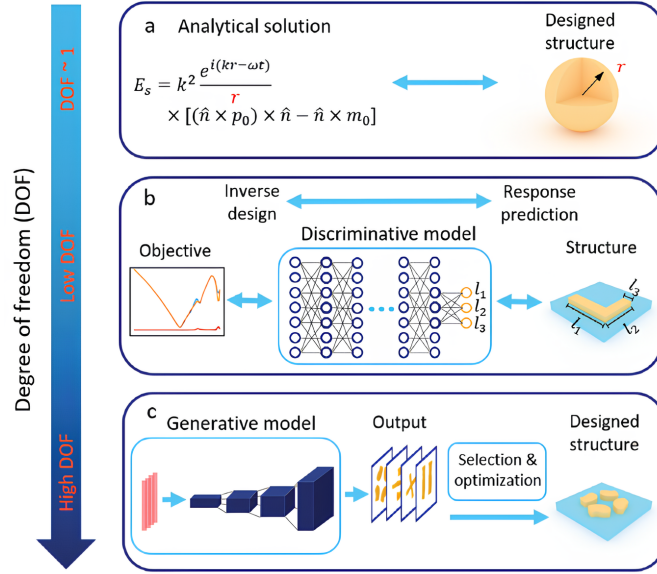


Figure 1: Machine learning algorithm for photonic design

To overcome traditional problems, a dataset using COMSOL was generated. This dataset included input parameters such as waveguide width, height, and gap, and corresponding outputs like effective index and coupling length. Chugh et al. then trained an artificial neural network (ANN) with these data to understand the relationship between inputs and outputs. Once trained, the model could make fast predictions — in just milliseconds — for new device configurations, without requiring any new simulation. Unlike FEM, which requires 2–3 minutes per data point, ML models can make predictions in under a second once trained. The Table 1 shows the hyperparameters of the neural network that the authors used in their implementation.

Results showed that the machine learning approach could accurately predict waveguide parameters—2 percent error in predicting effective index, 1–4 percent error for coupling length, and 5 percent error for power confinement — much faster than traditional simulation methods. Also, in Figure 2 the relation between the number of nodes, hidden layers, and MSE is shown.

Table 1: Hyperparameters of the neural network.

Activation Function	ReLU
Optimizer	Adam
Epochs	30,000
Loss Function	Mean Squared Error (MSE)
Number of Neurons per Layer	50

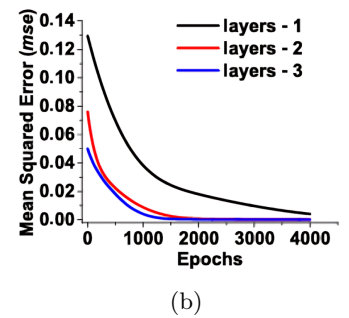
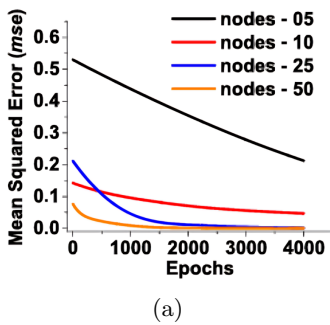


Figure 2: Mean squared error (MSE) using the training dataset for (a) different numbers of nodes with 2 hidden layers and (b) different numbers of hidden layers with 50 nodes in each hidden layer.

Although the use of machine learning significantly accelerated the prediction of photonic waveguide param-

eters, the model’s accuracy was entirely dependent on the quality and coverage of simulation data generated using COMSOL. For example, when training data were sparse in the 200–250 nm waveguide width range, the model exhibited high errors in predicting power confinement, indicating underfitting. Conversely, over-reliance on narrow or repetitive data during training could lead to overfitting, where the model fails to generalize to new waveguide designs.

Another drawback is that ML models, unlike traditional methods like FEM, do not offer physical interpretability and act as black boxes. If the model’s prediction is incorrect, it’s difficult to understand the underlying cause or trace it back to a specific physical behavior.

2.3 Deep Learning for Inverse Design of Power Splitters [4]

In this work, a deep learning-based inverse design framework for integrated photonic power splitters was presented. The method uses a deep residual neural network (ResNet) trained on 20,000 FDTD (Finite-Difference Time-Domain) simulations to instantly generate nanophotonic structures that match a target power splitting ratio with over 90 percent transmission efficiency and low reflection. The trained DNN eliminates the need for expensive trial-and-error FDTD simulations in the design loop. Same as the previous paper, the advantage of using deep learning for inverse design of power splitter is to save time for configuration.

Despite the impressive performance of the proposed DNN-based inverse design method, approximately 20,000 3D FDTD simulations were required to construct the dataset—each simulation taking several minutes—which in total consumed around two weeks on a high-end desktop with a Core i7 CPU and 64 GB RAM. To accelerate the process in more realistic scenarios, access to parallel computing resources and high-performance GPUs becomes essential. Moreover, training deep neural networks such as ResNet architectures—especially with 8 or more layers and wide hidden layers—also demands significant GPU memory. In Figure 3 the Network architectures that used are shown.



Figure 3: Mean squared error (MSE) using the training dataset for (a) different numbers of nodes with 2 hidden layers and (b) different numbers of hidden layers with 50 nodes in each hidden layer.

3. Conclusion

ML empowers photonic design by reducing simulation cost and enabling direct inverse design. However, reliance on simulated data, and lack of physical constraints are common drawbacks. Future work should prioritize developing physics-aware and data-efficient machine learning strategies, potentially coupled with faster simulation techniques, to overcome limitations and enable the discovery of globally optimal structures.

References

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